



BRIGHT MIND TECH

SPECIFICATIONS



FREE PANDAS

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I. PRESENTATION

1.1. Free Panda's Application Overview

In an increasingly interconnected world, managing and sharing online content is essential to foster discussion and collaboration. Free Panda's addresses this need by providing an integrated platform to capture, comment on, and share content, while ensuring the security and moderation needed to maintain a healthy and engaged community.

The "Free Panda's" application aims to provide a dynamic community platform for users to share and interact around content captured on the Internet. The project aims to create an environment where members can publish, comment, and exchange ideas while offering advanced management tools for administrators.

Free Panda's simplifies content management by allowing users to easily capture and organize information from various sources into one central platform. It supports diverse formats such as text, images, and videos, accommodating different communication styles and enhancing user interaction. This streamlined approach makes it easier for users to focus on engaging with content and participating in discussions rather than dealing with technical complexities.

The platform is built to foster a vibrant community where users can publish their own content, start discussions, and exchange ideas. Reflecting its name, Free Panda's emphasizes openness and inclusivity, encouraging users to express their opinions freely. Integrated features like commenting and engagement tracking enhance interaction, helping to build a sense of community and promote active collaboration.

Free Panda's includes advanced tools and robust security measures, including encryption and secure authentication, protect user data and maintain platform integrity. Its combination of user-friendly features, community engagement tools, and strong security measures makes it an essential platform for navigating today's digital landscape.

1.2. Presentation of the Company and Stakeholders

Company: Bright Mind Tech

- Role: Application development, project management, technical support.
- Responsibilities: Design, development, testing, deployment, and maintenance of the application.

Stakeholders:

Client: EPITECH CODING ACADEMY

- Role: Provide requirements, validate deliverables, and accept the final product.

Users:

- End Users: Members of the community using the application.
- Administrators: Platform managers, responsible for moderation and operations.

II. PROJECT DESCRIPTION

2.1. Specific Objectives and Expected Results

- Create an intuitive user interface for managing posts and comments.
- Develop social interaction features such as likes, favorites, and messaging.
- Set up management tools for administrators to monitor activities and maintain security.
- Ensure a smooth integration between the frontend (Vue.js) and the backend (Laravel) via a RESTful API.

2.2 Project Scope

Included:

- Frontend development in Vue.js.
- Backend development in Laravel.
- Integration of social authentication (Google, Facebook, LinkedIn).
- Implementation of a two-factor authentication system.
- Development of post, comment, notification, and favorite management features.

Excluded:

- Development of native mobile applications (iOS/Android).
- Integration of external features not specified in the specifications.

2.3 Context and Constraints

Context: The application will be deployed online and accessible via modern web browsers.

Constraints:

- Budget: Limited to the allocated resources.
- Time: Development deadlines to be respected for the launch of the product.
- Regulations: Compliance with data protection and privacy laws.

III. FUNCTIONALS REQUIREMENTS

3.1 Required Features

- Post Creation: Allow users to publish content with links, images, and descriptions.
- Comments and Replies: Comment functionality with response hierarchy.
- Likes: Post and comment rating system.
- Dashboards:
 - Admin: User, content, and statistics management.

- User: Personalized view of account activities and settings.
- User Rating: 5-stars rating system.
- Tag Management: Content creation and filtering by tags.
- Search: Search for posts and users.
- Account Verification: validation process by email.
- Social authentication and double authentication: login via social networks and protection by double authentication.
- Messaging: Send and receive private messages.
- Favorites: Add posts and users to favorites.
- Link Copy: Feature to copy links from posts.
- Notifications: Manage alerts and notifications.

3.2 Description of mains Processes

3.2.1. Post Creation

Access to the Feature:

The user logs into his account and accesses the "Create a Post" section via the main menu or the dashboard.

Entering Information:

- The user enters a source link (optional).
- The user uploads an image or captures an image via the dedicated interface.
- The user enters a description of the content.

Validation:

- The system checks the validity of the source link (if provided).
- The system validates the format and size of the image.
- The system checks the length of the description and others security verification.

Submission:

- The user submits the post via a confirmation button.
- The system saves the information in the database and makes the post visible on the platform.

Display:

The post is displayed on the news feed or in the appropriate section of the platform.

3.2.2. Comments and Replies

Accessing a Post:

User views a post from their news feed or via search.

Adding a Comment:

- User enters a comment in the field provided below the post.
- User submits the comment.

Replies to Comments:

- User selects the "Reply" button below an existing comment.
- A reply field opens below the selected comment.
- User enters and submits their reply.

Hierarchical Display:

Comments and replies are displayed hierarchically, with replies nested below the comments they reply to.

3.2.3. Likes

Interacting with a Post or Comment:

- User clicks the "Like" button below a post or comment.
- Updating the Counter:
- The system updates the likes counter for the post or comment in real time.

Display:

The number of likes is displayed and updated on the user interface.

3.2.4. Dashboards

Admin Dashboard:

- Access: The administrator logs in via a specific link to the admin interface.
- User Management: The administrator views user lists, adds, deletes, or edits accounts.
- Post and Comment Management: The administrator views and moderates posts and comments, with options to delete or edit them.
- Statistics: The administrator accesses key statistics such as the number of users, posts, number of likes, etc.

User Dashboard:

- Access: The user logs in to their account and accesses their personal dashboard.
- Custom View: The user views their favorite posts, notifications, and account settings.
- Favorites Management: The user views and manages their favorite posts and users.

3.2.5. User Rating

Access User Profile:

User views another user's profile

Enter Rating:

User selects a star rating from 1 to 5 and submits the rating.

Display Rating:

The user's average rating is calculated and displayed on their profile.

3.2.6. Tag Management

Create and Assign Tags:

- When creating a post, the user adds relevant tags.
- Tags are saved and associated with the post in the database.

Filter Content:

Users use tags to filter and display posts associated with those tags.

Tag Management:

Administrators add, edit, or delete tags via a dedicated interface.

3.2.7. Searching Users and Posts

Entering Search Criteria:

The user enters keywords or selects tags in the search bar.

Executing Search:

The system searches the database based on the criteria provided.

Displaying Results:

The search results are displayed as a list, sorted by relevance.

3.2.8. Account Verification

Sending Verification Link:

- After registration, an email with a verification link or code is sent.
- The user clicks on the link or enters the code to confirm their email address.

Account Activation:

Once verification is complete, the account is activated and the user can log in.

3.2.9. Social Authentication and Two-Factor Authentication

Social Authentication:

- User chooses to log in via Google, Facebook, or LinkedIn.
- User is redirected to the social authentication platform to enter their credentials.
- System receives an authentication token and creates a user account or logs the user in.

Two-Factor Authentication (2FA):

- User sets up two-factor authentication via an email.
- When logging in, user enters an additional code sent to the mail.

3.2.10. Messaging Between Users

Sending a Message:

- User accesses the inbox or composes a new message.
- User types the message and sends it.

Receiving Messages:

- Received messages are displayed in the user's inbox.
- Notifications are sent to inform the user of new messages.

3.2.11. Favorites (Posts and Users)

Add to Favorites:

User selects a post or user and clicks the "Add to Favorites" button.

Managing Favorites:

Favorite posts and users are visible in the appropriate section of the user dashboard.

Display:

Favorite posts and users are displayed with options to remove them from favorites.

IV. TECHNOLOGIES

Laravel

- Description: PHP framework for backend development.
- Features:
 - MVC Architecture: Clear separation of concerns between model, view, and controller.
 - Eloquent ORM: Database management with an elegant and easy-to-use interface.
 - Authentication and Security: Integration of authentication, verification, and security systems.

- Routing and Middleware: Management of routes and security layers for HTTP requests.

Vue.js

- Description: JavaScript framework for frontend development.
- Features:
 - Reactivity: Responsive and dynamic user interface with a data binding system.
 - Components: Creation of reusable components for better modularity.
 - Vue Router: Route management for single-page applications (SPAs).
 - Pinia: State management for centralized data management.

MySQL

- Description: Relational database management systems.
- Features:
 - MySQL: Fast and efficient database, suitable for web applications.
 - PostgreSQL: Powerful database with advanced features for complex queries and geospatial data.

RESTful API

- Description: Architecture for communication between the frontend and the backend.
- Features:
 - Endpoints: Defined entry points for CRUD operations (Create, Read, Update, Delete).
 - JSON: Data format for exchanges between the client and the server.
 - Authentication: Use of tokens to secure API requests.

OAuth for Authentication

- Description: Authorization protocol for authentication via third-party services.
- Features:
 - Google, Facebook, LinkedIn: Integration to allow users to log in with their social accounts.
 - Security: Management of access and refresh tokens to secure user sessions.

V. SECURITY PROTOCOLS

- SSL/TLS: Encryption of communications between the client and the server.
- Password Hashing: Use of bcrypt to secure passwords.
- Attack Prevention: Protection against CSRF, and SQL Injection attacks.

- Permission Management: Role-based access control (admin/user).

VI. PROJECT MANAGEMENT AND DEADLINES

Project Phases

- Planning (2 days): Definition of objectives, requirements, and resources.
- Development (6 days):
 - Backend (6 days): Development of features and APIs.
 - Frontend (6 days): Development of the user interface and integration with the backend.
 - Testing (1 day): Unit testing, integration testing, and performance testing.
- Deployment (1 day): Implementation and configuration of environments.
- Maintenance (Continuing): Support, updates, and bug fixes.

Intermediate Deliverables

- Mockups: Presentation of designs.
- Functional Prototypes: Preliminary versions for testing and feedback.
- Documentation: Technical and user documentation.